

1. Given the following for a Plane travelling from LL to Kolkata

$$\text{Mass of Plane (M)} = 10,000 \text{ kg}$$

$$C_L = 1.1$$

$$\text{Surface Area} = 40 \text{ m}^2$$

$$\rho = 1.22 \text{ kg/m}^3$$

$$\text{Distance (D)} = 4325 \text{ km}$$

a) Calculate speed of the airplane
first we need to calculate the lift force then we can calculate the velocity of the plane.

$$L = \frac{(C_L \times \rho \times S \times V^2)}{2} = \frac{\sqrt{2(98100 \times)}}{1.1 \times 1.22 \text{ kg/m}^3 \times 40 \text{ m}^2}$$

$$2L = C_L \times \rho \times S \times V^2 = \sqrt{196200 \text{ N} / 53.68 \text{ kg/m}^3}$$

$$V^2 = 2L / C_L \times \rho \times S$$

$$V = \sqrt{2L / (C_L \times \rho \times S)}$$

$$= \sqrt{3654.99 \text{ m/s}}$$

$$\begin{aligned} \text{Lift force} &= m \times g \\ &= 10000 \times 9.81 \text{ m/s}^2 \\ &= 98100 \text{ N} \end{aligned}$$

$$= \underline{\underline{60.5 \text{ m/s} \text{ ans.}}}$$

$$V = \sqrt{2L / (C_L \times \rho \times S)} \quad \nearrow$$

b) How long will it take to cover 4325 km @ 60.5 m/s

$$\text{Time} = \text{Distance} / \text{Speed}$$

$$T = D / S$$

$$T = \frac{4325000}{60.5 \text{ m/s}}$$

$$= \underline{\underline{20.02 \text{ Hrs}}} \text{ ans.}$$

c) Time it will take if the aircraft is being faced by 8 m/s headwind

$$\text{Speed} = 60.5 \text{ m/s} - 8 \text{ m/s} = 52.5 \text{ m/s}$$

$$\text{Distance} = 4325000 \text{ m}$$

$$\text{Time} = \frac{4325000}{52.5 \text{ m/s}}$$

$$= \underline{\underline{22.9 \text{ Hrs}}} \text{ ans.}$$

Ernest Sebastian.

Correction - Fixed Wing Aircraft - Lift & Drag.

2. After refuelling the plane, its heading to Sweden.

$$\text{Mass} : 12,500 \text{ kg}$$

$$\text{F. Area} : 17.5 \text{ m}^2$$

$$\text{S. Area} : 40 \text{ m}^2$$

$$\text{Density} : 1.22 \text{ kg/m}^3$$

$$\text{Distance} : 6175 \text{ km.}$$

a) Calculate the speed.

$$\begin{aligned} \text{Weight} &= 12,500 \times 9.81 \text{ kg} \\ &= 122525 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Velocity} &= \left(\frac{2 \times 122525}{1.22 \text{ kg} \times 40 \text{ m}^2 \times 1.1} \right) \\ &= \underline{\underline{67.6 \text{ m/s}}} \end{aligned}$$

$$\text{b) Drag Coefficient} = 0.15$$

$$\text{F. Area} = 17.5 \text{ m}^2$$

$$\begin{aligned} \text{Thrust} &= \frac{1}{2} \times 1.22 \times 67.6^2 \times 17.5 \times 0.15 \\ &= \underline{\underline{7315.7 \text{ N}}} \end{aligned}$$

c) Calculate Power

$$\begin{aligned} P &= \text{Thrust} \times \text{Velocity} \\ &= 7315.7 \times 67.6 \text{ m/s} \\ &= \underline{\underline{494484.9 \text{ W}}} \end{aligned}$$

d) Thrust produced with additional fuel tanks.

$$\text{Initial } T = \frac{1}{2} \times 1.22 \times V^2 \times 16.5 \times 0.125 \\ = 5748.0 \text{ N}$$

$$\text{Overall Thrust} = 7315.7 \text{ N} - 5748 \text{ N} \\ = \underline{\underline{1567 \text{ N}}}$$

e) Take off Mass for both journeys

MW - GH

$$\text{Thrust} = 5748 \text{ N}$$

$$1 \text{ L fuel} = 2000 \text{ N}$$

$$= 2.299 \text{ min}$$

$$\text{Time} = 1192.9 \text{ min}$$

GH/A $\rightarrow$ SWE

$$= 2.926 \text{ min}$$

$$\text{Time} = \underline{6175000}$$

$$67.6$$

$$= 1522.6$$

$$\text{Total mass} = (1522.6 \times 2.926) + (1522.6 + 2.299) \\ = 4455.1276 + 3500 \\ = \underline{\underline{7955.3 \text{ kg}}}$$